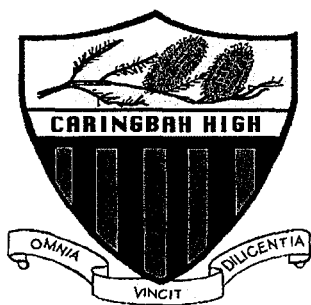




CARINGBAH HIGH SCHOOL

2014

**TRIAL HIGHER SCHOOL
CERTIFICATE EXAMINATION**

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11–16, show relevant mathematical reasoning and/or calculations

Total marks – 100

Section I Pages 2–5
10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II Pages 6–12
90 marks

- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section

SECTION 1 (10 marks)

Attempt Questions 1 - 10

Allow about 15 minutes for section.

Use the multiple choice answer sheet for questions 1 - 10

1. If w is a non-real cube root of unity the value $\frac{1}{1+w} + \frac{1}{1+w^2}$ is equal to

- (A) -1 (B) 0 (C) 1 (D) none of these

2. What is the remainder when $x^3 + x^2 + 5x + 6$ is divided by $x + i$

- (A) $7 - 4i$ (B) $7 - 6i$ (C) $5 - 4i$ (D) $5 + 6i$

3. The gradient of the tangent to $xy^3 + 2y = 4$ at the point $(2, 1)$ is

- (A) -8 (B) $\frac{1}{8}$ (C) 8 (D) $-\frac{1}{8}$

4. The eccentricity of the ellipse $3x^2 + 5y^2 - 15 = 0$ is

- (A) $\sqrt{\frac{5}{2}}$ (B) $\sqrt{\frac{2}{5}}$ (C) $\sqrt{\frac{8}{5}}$ (D) $\sqrt{\frac{5}{8}}$

5. The polynomial $3x^3 - 2x^2 + x - 7 = 0$ has roots α, β, γ .

Which polynomial has roots $\frac{2}{\alpha}, \frac{2}{\beta}, \frac{2}{\gamma}$?

- (A) $3x^3 - 4x^2 + 4x - 56$ (B) $7x^3 - 2x^2 + 8x - 24 = 0$
(C) $9x^3 - 2x^2 - 27x - 49 = 0$ (D) $24x^3 - 8x^2 + 2x - 7 = 0$

6. The arg of iz where $z = 1 + i$ is

- (A) $-\frac{\pi}{4}$ (B) $\frac{3\pi}{4}$ (C) $\frac{5\pi}{4}$ (D) $-\frac{3\pi}{4}$

7. Find $\int x \sin(x^2 + 3) dx$

- (A) $-\frac{1}{2} \cos(x^2 + 3) + C$ (B) $-\frac{1}{2} \sin(x^2 + 3) + C$
 (C) $\frac{1}{2} \cos(x^2 + 3) + C$ (D) $2x \cos(x^2 + 3) + C$

8. The polynomial equation $P(x) = 0$ has real coefficients, and has roots which include

$$x = -2 + i \quad \text{and} \quad x = 2.$$

What is the minimum possible degree of $P(x)$?

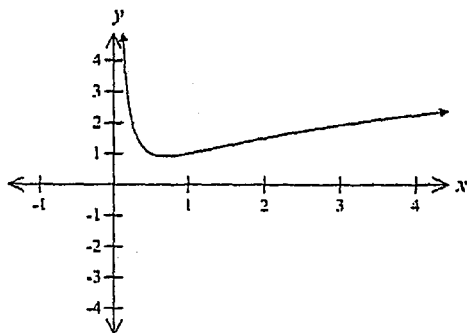
- (A) 1 (B) 2 (C) 3 (D) 4

9. What is the value of $\int_0^1 \frac{e^x}{1+e^x} dx$

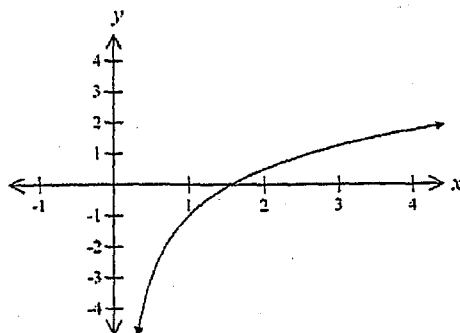
- (A) $\log_e(1+e)$ (B) 1
 (C) $\log_e\left(\frac{1+e}{2}\right)$ (D) $\log_e \frac{e}{2} - 2$

10. Which of the following is the sketch of $y = \log_2 x + \frac{1}{x}$?

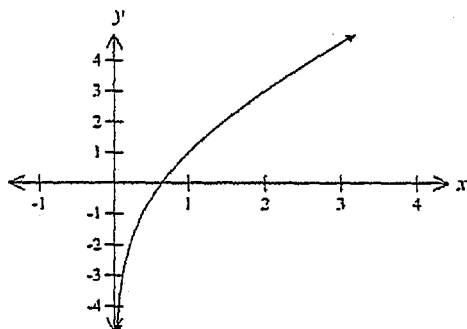
(A)



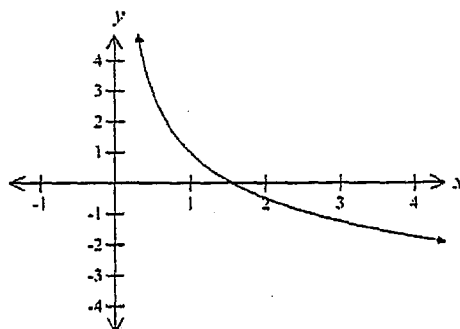
(B)



(C)



(D)



Question 11 (15 marks)

a) Let $w = \sqrt{3} + i$ and $z = 3 - \sqrt{3}i$

i) Find wz (1)

ii) Express w in mod/arg form (2)

iii) Write w^4 in simplest Cartesian form. (2)

b)

i) Mark clearly on an Argand diagram the region satisfied simultaneously by (2)

$$|z + 2| < 2 \quad \text{and} \quad 0 < \arg z < \frac{3\pi}{4}$$

ii) Solve simultaneously (2)

$$|z + 2| = 2 \quad \text{and} \quad \arg z = \frac{3\pi}{4}$$

Write your answer in the form $a + ib$

c) A polynomial $P(x)$ has a double root at $x = \alpha$, ie $P(x) = (x - \alpha)^2 Q(x)$

i) Prove that $P'(x)$ also has a root at $x = \alpha$ (2)

ii) The polynomial $Q(x) = x^4 - 6x^3 + ax^2 + bx + 36$ has a double root at $x = 3$ (2)

Find the values of a and b

iii) Factorise $Q(x)$ over the complex field. (2)

Question 12 (15 marks)

a)

i) Show that $(\cos x - \sin x)^2 = 1 - \sin 2x$ (1)

ii) Evaluate (2)

$$\int_0^{\frac{\pi}{4}} \sqrt{1 - \sin 2x} \, dx$$

b) Using the substitution $u = 1 - x$, find (3)

$$\int x\sqrt{1-x} \, dx$$

c)

i) Find the value of the integral (2)

$$\int_0^{\pi} \frac{1}{\sqrt{16-x^2}} \, dx$$

ii) Find the integral of (2)

$$\int \frac{1}{16-x^2} \, dx$$

d) If $I_n = \int_1^e x(\ln x)^n \, dx$, $n = 0, 1, 2, 3, \dots$

i) Show that $I_n = \frac{e^2}{2} - \frac{n}{2} I_{n-1}$, $n=1, 2, 3, \dots$ (3)

ii) Hence evaluate (2)

$$\int_1^e x(\ln x)^3 \, dx$$

Question 13 (15 marks)

a)

i) Show that the equation of the normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at the point $P(a \sec \theta, b \tan \theta)$ is $ax \sin \theta + by = (a^2 + b^2) \tan \theta$ (3)

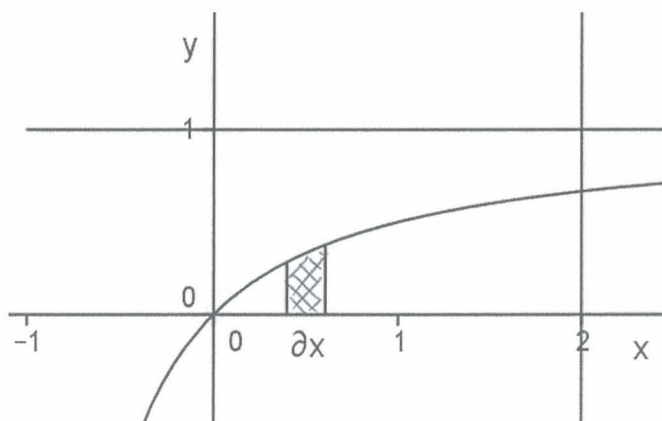
ii) If the normal in part (i) intersects the x axis at A and the y axis at B, find the co-ordinates of A and B. (2)

iii) Show that the co-ordinates of M, the midpoint of AB are given by $x = \frac{1}{2a}(a^2 + b^2) \sec \theta$, $y = \frac{1}{2b}(a^2 + b^2) \tan \theta$ (2)

iv) Hence find the equation of the locus of M in Cartesian form. (2)

v) If $a = b$, what can you say about the locus in part (iv) (1)

b) The region bounded by the portion of the curve $y = \frac{x}{x+1}$, and the x axis is rotated about the line $x = 2$



i) Using the method of cylindrical shells, show that the volume δV of a typical shell at a distance x from the origin and with thickness δx is given by (1)

$$\delta V = 2\pi(2 - x) \cdot \frac{x}{1 + x} \cdot \delta x$$

ii) Hence find the volume of this solid. (4)

Question 14 (15 marks)

- a) Consider the function $f(x) = (3 - x)(x + 1)$ on separate axes sketch, showing the important features the graphs of

i) $y = f(x)$ (1)

ii) $y = |f(x)|$ (1)

iii) $y = f(|x|)$ (1)

iv) $|y| = f(x)$ (1)

v) $y^2 = f(x)^3$ (2)

- b) Given $a + b = m$, prove that, for $a > 0$, $b > 0$, and $m > 0$

i) $\frac{1}{a} + \frac{1}{b} \geq \frac{4}{m}$ (2)

ii) $\frac{1}{a^2} + \frac{1}{b^2} \geq \frac{8}{m^2}$ (2)

c) Evaluate $\lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta}$ (2)

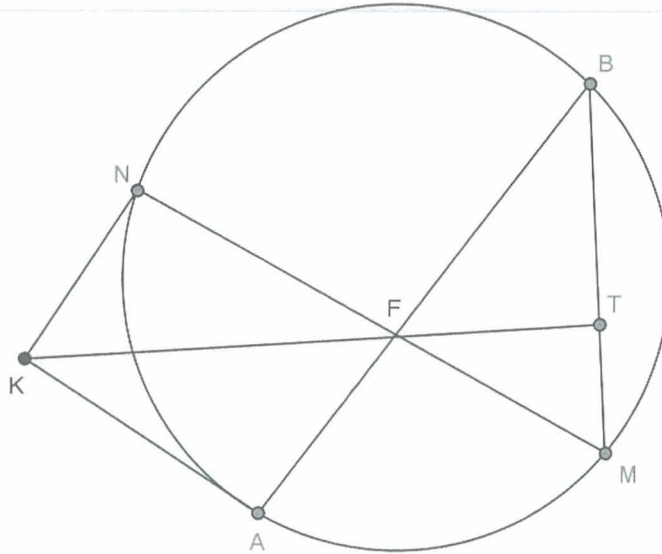
d)

i) Use De Moivre's Theorem to express $\cos 3\theta$ and $\sin 3\theta$ in terms of powers of $\sin \theta$ and $\cos \theta$ (2)

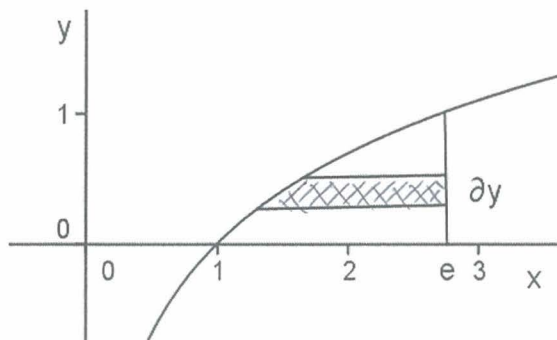
ii) Hence express $\tan 3\theta$ as a rational function of t , where $t = \tan \theta$ (1)

Question 15 (15 marks)

- a) As shown below, a circle has two chords AB and MN intersecting at F . Perpendiculars are drawn to these chords at A and at N , intersecting at K . KF produced, meets MB at T . Prove that KT is perpendicular to MB (Hint: Join AN and let $\angle ANF = \theta^\circ$) (4)



- b) If $V_1 = 1$, $V_2 = 5$ and $V_n = 5V_{n-1} - 6V_{n-2}$ for $n \geq 3$, show that $V_n = 3^n - 2^n$ for $n \geq 1$ (3)
- c) Consider the curve $y = \ln x$ sketched below.



- Use the method of slicing to find the volume obtained by rotating the region bounded by $1 \leq x \leq e$, $0 \leq y \leq \ln x$, about the y axis. (3)

Question 15 (Cont'd)

d) The equation $x^3 - 3x^2 - x + 2 = 0$ has roots α, β, γ . Find equations with roots

i) $2\alpha + \beta + \gamma, \alpha + 2\beta + \gamma, \alpha + \beta + 2\gamma$ (3)

ii) Find the value of the sum of the squares of the roots of the equation formed in (i) (2)

Question 16 (15 marks)

a) Find $\int \sin^5 \theta \cos^4 \theta d\theta$ (3)

b) If $x^2 + y^2 + xy = 3$

i) Find $\frac{dy}{dx}$ (2)

ii) Sketch showing critical points and stationary points the graph of $x^2 + y^2 + xy = 3$ (3)

c)

i) If $x_1 > 1$ and $x_2 > 1$ show that $x_1 + x_2 > \sqrt{x_1 x_2}$ (3)

ii) Use the Principal of Mathematical Induction to show that, (4)
For $n \geq 2$, if $x_j > 1$ where $j = 1, 2, 3, \dots, n$ then,

$$\ln(x_1 + x_2 + \dots + x_n) > \frac{1}{2^{n-1}} (\ln x_1 + \ln x_2 + \dots + \ln x_n)$$

END OF PAPER

Find $\int x \sin(x^2 + 3) dx$

(A) $-\frac{1}{2} \cos(x^2 + 3) + c$

(B) $-\frac{1}{2} \sin(x^2 + 3) + c$

(C) $\frac{1}{2} \cos(x^2 + 3) + c$

(D) $2x \cos(x^2 + 3) + c$

If ω is a non-real cube root of unity the value of $\frac{1}{1+\omega} + \frac{1}{1+\omega^2}$ is equal to

(a) -1

(B) 0

(C) 1

(D) None of these

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Multiple Choice

C 2, C 3, D 4, B 5, B
B 7, A 8, C 9, C 10, A

Question 11

1 (i) WZ

$$= (\sqrt{3}+i)(3-\sqrt{3}i) \\ = 3\sqrt{3} - 3i + 3i + \sqrt{3} \\ = 4\sqrt{3}$$

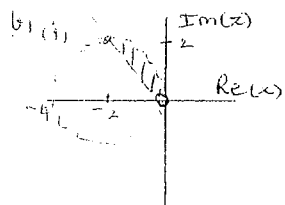
$$1. r = \sqrt{(\sqrt{3})^2 + (1)^2} \\ = 2$$

$$\tan \theta = \frac{1}{\sqrt{3}}$$

$$\theta = \pi/6$$

$$w = 2 \operatorname{cis} \pi/6$$

$$(i) w^4 = (2 \operatorname{cis} \pi/6)^4 \\ = 2^4 \operatorname{cis} 4\pi/6 \\ = -8 + 8\sqrt{3}i$$



$$(ii) A(-2+2i)$$

$$(c) (i) P(x) = (x-\alpha)^2 Q(x)$$

$$P'(x) = 2(x-\alpha)Q(x) + (x-\alpha)^2 Q'(x) \\ = (x-\alpha)[2Q(x) + (x-\alpha)Q'(x)]$$

$\therefore P'(x)$ has a factor $(x-\alpha)$

$\therefore x = \alpha$ is a root

$$(c) (ii) P(x) = x^4 - 6x^3 + ax^2 + bx + 36$$

$$P(3) = 0$$

$$0 = 81 - 162 + 9a + 3b + 36$$

$$3a + b = 15 \quad \text{--- (1)}$$

$$P'(3) = 0$$

$$P'(x) = 4x^3 - 18x^2 + 2ax + b$$

$$0 = 4(27) - 18(9) + 6a + b$$

$$6a + b = 54 \quad \text{--- (2)}$$

$$\textcircled{1} - \textcircled{2} \quad -3a = -39$$

$$a = 13, b = -24$$

$$(c) (iii) P(x) = x^4 - 6x^3 + 13x^2 - 24x + 36$$

$$P(x) = (x-3)^2(x^2+mx+n)$$

$$= (x^2-6x+9)(x^2+mx+n)$$

$$= x^4 + x^3(m-6) + x^2(-6m+n+9) + x(-6n+9m) + 9n$$

$$\therefore 9n = 36 \quad -6m+n+9 = 13$$

$$n = 4 \quad -6m + 13 = 13 \\ m = 0$$

Factors of $P(x)$

$$P(x) = (x-3)^2(x^2+4)$$

$$= (x-3)^2(x+2i)(x-2i)$$

$$\text{OR } x^2 - 6x + 9 = \frac{x^2 + 4}{\frac{x^4 - 6x^3 + 13x^2 - 24x + 36}{x^2 - 6x + 9}} \\ \frac{4x^2 - 24x + 36}{4x^2 - 24x + 36}$$

$$\therefore P(x) = (x-3)^2(x^2+4)$$

$$= (x-3)^2(x-2i)(x+2i)$$

Question 12

$$(a) (i) (\cos x - \sin x)^2 = 1 - \sin 2x$$

L.H.S.

$$\cos^2 x - 2\sin x \cos x + \sin^2 x \\ = \cos^2 x + \sin^2 x - 2\sin x \cos x \\ = 1 - 2\sin x \cos x$$

$$= 1 - \sin 2x$$

$$(ii) \int_0^{\pi/4} \sqrt{1 - \sin 2x} \, dx$$

$$= \int_0^{\pi/4} \cos x - \sin x \, dx$$

$$= [\sin x + \cos x]_0^{\pi/4}$$

$$= \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\right) - (1)$$

$$= \sqrt{2} - 1$$

$$(b) \int x \sqrt{1-x} \, dx \quad \text{Let } u = 1-x$$

$$x = 1-u$$

$$\frac{du}{dx} = -1$$

$$dx = -du$$

$$= \int (1-u)u^{1/2} \, -du$$

$$= \int (1-u)u^{1/2} \, du$$

$$= \int u^{3/2} - u^{1/2} \, du$$

$$= \frac{2}{5}u^{5/2} - \frac{2}{3}u^{3/2} + C$$

$$= \frac{2}{5}(1-x)^{5/2} - \frac{2}{3}(1-x)^{3/2} + C$$

$$(c) (i) \int_0^{\pi/4} \frac{1}{\sqrt{16-x^2}} \, dx$$

$$= \int_0^{\pi/4} \frac{1}{\sqrt{4^2-x^2}} \, dx$$

$$= \left[\sin^{-1} \frac{x}{4} \right]_0^{\pi/4}$$

$$(c) (ii) \int \frac{1}{16-x^2} \, dx$$

$$= \frac{1}{8} \int \frac{1}{4-x} + \frac{1}{4+x} \, dx$$

$$= \frac{1}{8} [-\ln|4-x| + \ln|4+x|]$$

$$= \frac{1}{8} \ln \left| \frac{4+x}{4-x} \right| + C$$

$$\frac{1}{(4-x)(4+x)} = \frac{A}{4-x} + \frac{B}{4+x}$$

$$= 4A + 2A + 4B - 2B$$

$$(4-x)(4+x)$$

$$4A + 4B = 1 \quad A - B = 0$$

$$A = B$$

$$A = \frac{1}{8}, B = \frac{1}{8}$$

$$(d) I_n = \int_1^e x (\ln x)^n \, dx \quad n=0,1,2$$

$$\text{Let } u = (\ln x)^n, v' = \frac{1}{x} \\ v = \frac{x^2}{2}$$

$$I_n = uv - \int v \, du$$

$$= \left[(\ln x)^n \cdot \frac{x^2}{2} \right]_1^e - \int_1^e \frac{x^2}{2} n (\ln x)^{n-1} \cdot \frac{1}{x} \, dx$$

$$= \left[(\ln x)^n \cdot \frac{x^2}{2} \right]_1^e - \frac{n}{2} \int_1^e x (\ln x)^{n-1} \, dx$$

Question 12 (cont'd)

$$b) I_n = \left[(\ln e)^2 \cdot \frac{e^2}{2} \right] - \frac{n}{2} \int I_{n-1}$$

$$I_n = \frac{e^2}{2} - \frac{n}{2} I_{n-1}$$

$$i) \int_1^e x (\ln x)^3 dx$$

$$I_n = \frac{e^2}{2} - \frac{3}{2} \left[\int_1^e x (\ln x)^2 dx \right]$$

$$= \frac{e^2}{2} - \frac{3}{2} \left[\frac{e^2}{2} - I_1 \right]$$

$$= \frac{e^2}{2} - \frac{3}{2} \cdot \frac{e^2}{2} + \frac{3}{2} I_1$$

$$= \frac{e^2}{2} - \frac{3e^2}{4} + \frac{3}{2} \left[\frac{e^1}{2} - \frac{1}{2} e^0 \right]$$

$$= \frac{e^2}{2} - \frac{3e^2}{4} + \frac{3e^2}{4} - \frac{3}{4} \int_1^e x dx$$

$$= \frac{e^2}{2} - \frac{3}{4} \left[\frac{x^2}{2} \right]_1^e$$

$$= \frac{e^2}{2} - \frac{3}{4} \left[\frac{e^2}{2} - \frac{1}{2} \right]$$

$$= \frac{e^2}{2} - \frac{3e^2}{8} + \frac{3}{8}$$

$$= \frac{e^2 + 3}{8}$$

Question 13

$$(a) (i) \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\frac{dx}{dx} = \frac{b^2 x}{a^2 y}$$

$$\text{grad of normal} = -\frac{a^2 y}{b^2 x}$$

$$\text{at P} = -\frac{a^2 b \tan \theta}{b^2 a \sec \theta} = -\frac{a}{b} \sin \theta$$

$\therefore$ Eqn of normal

$$y - b \tan \theta = -\frac{a}{b} \sin \theta (x - a \sec \theta)$$

$$by - b^2 \tan \theta = -ax \sin \theta + a^2 \sin \theta \sec \theta$$

$$ax \sin \theta + by = (a^2 + b^2) \tan \theta$$

(ii) At A, $y=0$

$$x = \frac{a^2 + b^2}{a \sin \theta} \tan \theta$$

$$= \frac{a^2 + b^2}{a} \sec \theta$$

At B, $x=0$

$$y = \frac{a^2 + b^2}{b} \tan \theta$$

$$(ii) x = \frac{1}{2} \left[\frac{a^2 + b^2}{a} \sec \theta + 0 \right] = \frac{a^2 + b^2}{2a} \sec \theta \quad \text{--- (1)}$$

$$y = \frac{1}{2} \left[\frac{a^2 + b^2}{b} \tan \theta + 0 \right]$$

$$y = \frac{a^2 + b^2}{2b} \tan \theta \quad \text{--- (2)}$$

Question 13 (cont'd)

from (iii) (1) and (2)

$$(iv) \frac{2ax}{a^2 + b^2} = \sec \theta - \theta$$

$$\frac{2by}{a^2 + b^2} = \tan \theta - \phi$$

$$\theta^2 - \phi^2$$

$$= \frac{4a^2 x^2}{(a^2 + b^2)^2} - \frac{4b^2 y^2}{(a^2 + b^2)^2} = 1$$

$$4a^2 x^2 - 4b^2 y^2 = (a^2 + b^2)^2$$

(v) If $a=b$

$$4a^2 x^2 - 4a^2 y^2 = 4a^4$$

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

when $a=b$

$$x^2 - y^2 = a^2$$

Rectangular Hyperbola

$$(vi) \Delta V = \left[\pi (2-x)^2 - \pi [2-(x+a)]^2 \right] \cdot \frac{x}{x+1}$$

$$\Delta V = \pi \left\{ (2-x)^2 - [2-(x+a)]^2 \right\} \cdot \frac{x}{x+1}$$

$$= \pi \left\{ \Delta x [4-2x - \Delta x] \right\} \cdot \frac{x}{x+1}$$

$$= 2\pi \left\{ (2-x) \Delta x \right\} \cdot \frac{x}{x+1}$$

$$(b) V = 2\pi \int_0^2 (2-x) \frac{x}{x+1} dx$$

$$V = 2\pi \int_0^2 -x+3 - \frac{3}{x+1} dx$$

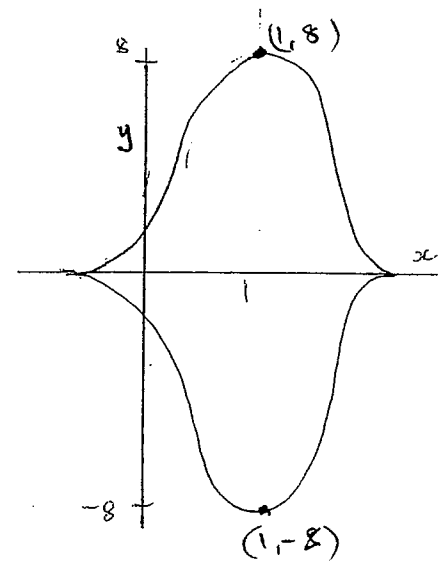
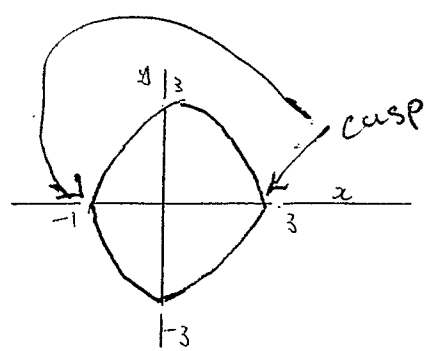
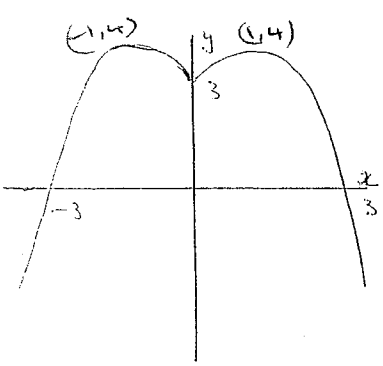
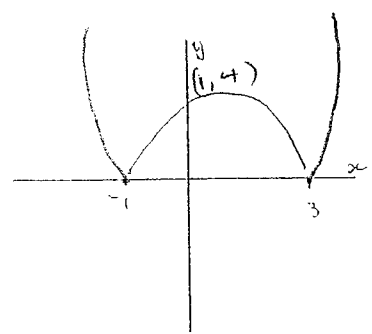
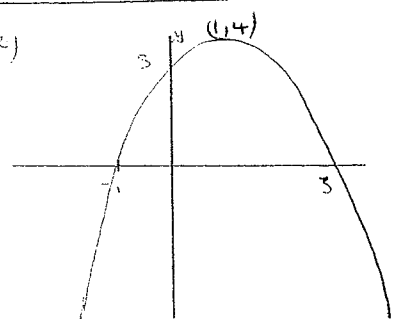
$$V = 2\pi \left[-\frac{1}{2}x^2 + 3x - 3 \ln(x+1) \right]_0^2$$

$$V = 2\pi \left[(-2+6-3 \ln 3) - 0 \right]$$

$$V = 2\pi \left[4 - 3 \ln 3 \right] u^3$$

Question 14

(a)



Question 14

(i) $\frac{1}{a} + \frac{1}{b} - \frac{4}{m} = \frac{1}{a} + \frac{1}{b} - \frac{4}{a+b}$

R.H.S. $\frac{1}{a} + \frac{1}{b} - \frac{4}{a+b}$
 $= \frac{b(a+b) + a(a+b) - 4ab}{ab(a+b)}$
 $= \frac{(a+b)^2 - 4ab}{ab(a+b)}$
 $= \frac{a^2 - 2ab + b^2}{ab(a+b)}$
 $= \frac{(a-b)^2}{ab(a+b)}$

now since $a > 0, b > 0, (a-b)^2 > 0$
 $ab(a+b) > 0$ for all a, b

Hence $\frac{1}{a} + \frac{1}{b} - \frac{4}{m} > 0$
 $\frac{1}{a} + \frac{1}{b} > \frac{4}{m}$

(ii) $x^2 + y^2 \geq 2xy$

Let $x = \frac{1}{a}, y = \frac{1}{b}$

$\frac{1}{a^2} + \frac{1}{b^2} \geq \frac{2}{ab}$ — ①

from (i) $\frac{1}{a} + \frac{1}{b} \geq \frac{4}{m}$

$\frac{a+b}{ab} \geq \frac{4}{m} \quad (a+b=m)$

$\frac{1}{ab} \geq \frac{4}{m^2}$

(i) from ①

$\frac{1}{a^2} + \frac{1}{b^2} \geq \frac{2}{\frac{m^2}{2}}$

$\frac{1}{a^2} + \frac{1}{b^2} \geq \frac{4}{m^2}$

(c) $\lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta}$

$= \frac{1 - \cos \theta}{\theta} \times \frac{1 + \cos \theta}{1 + \cos \theta}$

$= \frac{1 - \cos^2 \theta}{\theta(1 + \cos \theta)}$

$= \frac{\sin^2 \theta}{\theta(1 + \cos \theta)}$

$= \frac{\sin \theta}{\theta} \times \frac{\sin \theta}{1 + \cos \theta}$

$= 1 \times 0$

$= 0$

Question 14 cont'd

$$(i) (cis \theta)^3 = (\cos \theta + i \sin \theta)^3$$

$$\begin{aligned} cis \theta^3 &= \cos^3 \theta - 3\sin^2 \theta \cos \theta + 3i(\cos^2 \theta \sin \theta - \sin^3 \theta) \\ &= \cos^3 \theta - 3\sin^2 \theta \cos \theta + i(3\cos^2 \theta \sin \theta - \sin^3 \theta) \end{aligned}$$

Equate Real & Imaginary

$$\cos 3\theta = \cos^3 \theta - 3\sin^2 \theta \cos \theta \quad \text{--- (1)}$$

$$\sin 3\theta = 3\cos^2 \theta \sin \theta - \sin^3 \theta \quad \text{--- (2)}$$

$$\tan 3\theta = \frac{(2)}{(1)}$$

$$\tan 3\theta = \frac{3\cos^2 \theta \sin \theta - \sin^3 \theta}{\cos^3 \theta - 3\sin^2 \theta \cos \theta}$$

$\div$ Rhs top & bottom by $\cos^3 \theta$

$$= \frac{3 \frac{\sin \theta}{\cos \theta} - \frac{\sin^3 \theta}{\cos^3 \theta}}{1 - 3 \frac{\sin^2 \theta}{\cos^2 \theta}}$$

$$= \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$$

$$= \frac{3t - t^3}{1 - 3t^2}$$

where $t = \tan \theta$

(7)

Question 15

(a) $\angle ANM = \angle ABM = \theta$ (angles at circumference on same chord AM)

$$\angle KNF + \angle KAF = 180^\circ$$

$\therefore$ KNFA is a cyclic quad

$\angle AKF = \angle ANF = \theta$ (angles at circumference on same chord of circle through KNFA)

$$\angle KFA = 90 - \theta \text{ (angle sum } \triangle AKF)$$

$$\angle BFT = \angle KFA = 90 - \theta \text{ (vert opp } \angle \text{'s)}$$

$$\angle FTB = 180 - (\angle BFT + \angle FBM)$$

$$\angle ABM = \angle FBM \text{ (same angle)}$$

$$\begin{aligned} \angle FTB &= 180 - [(90 - \theta) + \theta] \\ &= 90^\circ \end{aligned}$$

$$\therefore KF \perp MB$$

$$(4) V_1 = 1, V_2 = 5, V_n = 3^n - 2^n, n \geq 1$$

$$V_n = 5V_{n-1} - 6V_{n-2}$$

$$\begin{aligned} V_1 &= 3^1 - 2^1 \\ &= 1 \end{aligned}$$

$$\begin{aligned} V_2 &= 3^2 - 2^2 \\ &= 5 \end{aligned}$$

Assume true for $n = k$

$$V_k = 3^k - 2^k \quad k \geq 1$$

(8)

Question 15 (cont'd)

prove true for $n=k+1$

$$\begin{aligned}
 k+1 &= 5V_{k+1-1} - 6V_{k+1-2} \\
 &= 5V_k - 6V_{k-1} \\
 &= 5[3^k - 2^k] - 6[3^{k-1} - 2^{k-1}] \\
 &= 5 \cdot 3^k - 5 \cdot 2^k - 6[2 \cdot 3^{k-1} - 3 \cdot 2^{k-1}] \\
 &= 5 \cdot 3^k - 5 \cdot 2^k - 2 \cdot 3^k + 3 \cdot 2^k \\
 &= 3 \cdot 3^k - 2 \cdot 2^k \\
 &= 3^{k+1} - 2^{k+1}
 \end{aligned}$$

plus statement or M.I. proof

$$\begin{aligned}
 I \int_V &= \int_0^1 [\pi e^2 - \pi (e^u)^2] dy \\
 \int_V &= \pi [e^2 - e^{2u}] dy \\
 V &= \pi \int_0^1 [e^2 - e^{2u}] dy \\
 V &= \pi \left[ye^2 - \frac{1}{2} e^{2u} \right]_0^1 \\
 V &= \pi \left[\left(e^2 - \frac{1}{2} e^2 \right) - \left(0 - \frac{1}{2} \right) \right] \\
 &= \pi \left[\frac{1}{2} e^2 + \frac{1}{2} \right] \\
 &= \frac{\pi}{2} [e^2 + 1] u^3.
 \end{aligned}$$

Question 15

(d) $x^3 - 3x^2 - x + 2 = 0$ — (1)

$\alpha + \beta + \gamma = 3.$

(i) $2\alpha + \beta + \gamma, \alpha + 2\beta + \gamma, \alpha + \beta + 2\gamma$

$x = \alpha + \beta + \gamma$

$x = \alpha + 3$

$\therefore \alpha = x - 3.$

Sub into (1)

$$\begin{aligned}
 (x-3)^3 - 3(x-3)^2 - (x-3) + 2 &= 0 \\
 x^3 - 12x^2 + 44x - 49 &= 0
 \end{aligned}$$

(ii) $x^2 + \beta^2 + \gamma^2$

$$\begin{aligned}
 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) \\
 &= \left(\frac{12}{1}\right)^2 - 2\left(\frac{44}{1}\right) \\
 &= 144 - 88 \\
 &= 56.
 \end{aligned}$$

Question 1b

$$\int \sin^5 \theta \cos^4 \theta d\theta$$

$$= \int \sin^4 \theta \cos^4 \theta (-\sin \theta) d\theta$$

$$= -\int \sin^4 \theta \cos^4 \theta d\theta + 2 \int \cos^6 \theta (-\sin \theta) d\theta - \int \cos^8 \theta (-\sin \theta) d\theta$$

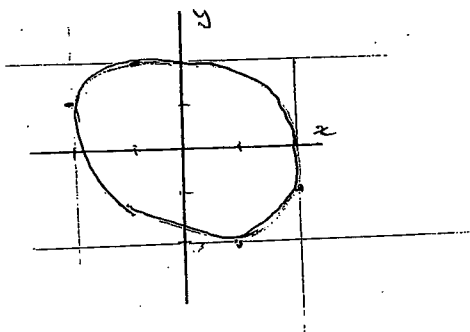
$$= -\frac{1}{5} \cos^5 \theta + \frac{2}{7} \cos^7 \theta - \frac{1}{9} \cos^9 \theta + C$$

$$x^2 + y^2 + xy = 3$$

$$\frac{dy}{dx} = -\frac{(2x+y)}{x+2y}$$

The tangents at the critical points $(-1, 2)$ and $(1, -2)$ are horizontal

The tangents at the critical points $(-2, 1)$ and $(2, -1)$ are vertical



(11)

Question 1b

(C) (i) $x_1 > 1, x_2 > 1$

Consider

$$(\sqrt{x_1} - \sqrt{x_2})^2 > 0$$

$$x_1 - 2\sqrt{x_1 x_2} + x_2 > 0$$

$$x_1 + x_2 > 2\sqrt{x_1 x_2}$$

$$x_1 + x_2 > \sqrt{x_1 x_2}$$

(C) (ii)

To prove $\ln(x_1 + x_2 + \dots + x_n) > \frac{1}{n-1} (\ln x_1 + \ln x_2 + \dots + \ln x_n)$
when $n=2$ we know

$$x_1 + x_2 > \sqrt{x_1 x_2}$$

$$\ln(x_1 + x_2) > \ln \sqrt{x_1 x_2} \text{ since } x_1, x_2 \text{ are both } > 1$$

$$\ln(x_1 + x_2) > \ln (x_1 x_2)^{1/2}$$

$$\ln(x_1 + x_2) > \frac{1}{2} \ln(x_1 x_2)$$

$$\ln(x_1 + x_2) > \frac{1}{2} (\ln x_1 + \ln x_2)$$

Assume true for $n \leq k$

$$\ln(x_1 + x_2 + \dots + x_k) > \frac{1}{k-1} (\ln x_1 + \ln x_2 + \dots + \ln x_k)$$

when $n = k+1$

$$\text{L.H.S. } \ln(x_1 + x_2 + \dots + x_k + x_{k+1}) > \frac{1}{2} [\ln(x_1 + x_2 + \dots + x_k) + \ln x_{k+1}]$$

$$\text{R.H.S. } \left[\frac{1}{2^{k-1}} < 1 \right]$$

$$> \frac{1}{2} \left[\frac{1}{2^{k-1}} (\ln x_1 + \ln x_2 + \dots + \ln x_k) + \ln x_{k+1} \right]$$

$$> \frac{1}{2} \left[\frac{1}{2^{k-1}} (\ln x_1 + \ln x_2 + \dots + \ln x_k + \frac{1}{2^{k-1}} \ln x_{k+1}) \right]$$

$$\ln(x_1 + x_2 + \dots + x_k + x_{k+1}) > \frac{1}{2^k} [\ln x_1 + \ln x_2 + \dots + \ln x_k + \ln x_{k+1}]$$

* Plus M.I statement

(12)

Multiple Choice.

Solutions

$$1/ \quad \frac{1}{1+\omega} + \frac{1}{1+\omega^2}$$

$$1+\omega+\omega^2=0$$

$$\omega = 1$$

$$= \frac{1+\omega^2 + 1+\omega}{(1+\omega)(1+\omega^2)}$$

$$= \frac{1}{1+\omega^2+\omega+\omega^3}$$

$$= \frac{1}{1}$$

$$= 1$$

(C)

$$2/ \quad P(x) = x^3 + x^2 + 5x + 6$$

$$P(-i) = (-i)^3 + (-i)^2 + 5(-i) + 6$$

$$= i - 1 - 5i + 6$$

$$= 5 - 4i$$

(C)

$$3/ \quad xy^3 + 2y = 4$$

$$y^3 \cdot 1 \cdot dx + x \cdot 3y^2 \cdot dy + 2 \cdot dy = 0$$

$$y^3 + 3xy^2 \frac{dy}{dx} + 2 \frac{dy}{dx} = 0$$

$$y^3 + \frac{dy}{dx} (3xy^2 + 2) = 0$$

$$\frac{dy}{dx} = \frac{-y^3}{3xy^2 + 2}$$

(D)

$$\text{when } x=2, y=1 \quad \frac{dy}{dx} = \frac{-1}{8}$$

$$4, \quad 3x^2 + 5y^2 - 15 = 0$$

$$\frac{x^2}{5} + \frac{y^2}{3} = 1$$

$$b^2 = a^2(1 - e^2)$$

$$\frac{3}{5} = 1 - e^2$$

$$e = \sqrt{2/5}$$

(B)

$$5, \quad \text{let } y = \frac{2}{x}$$

$$x = \frac{2}{y}$$

$$3\left(\frac{2}{y}\right)^3 - 2\left(\frac{2}{y}\right)^2 + \frac{2}{y} - 7 = 0$$

$$\frac{24}{y^3} - \frac{8}{y^2} + \frac{2}{y} - 7 = 0$$

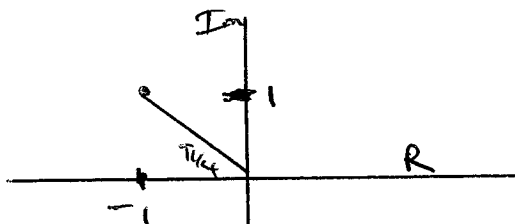
$$24 - 8y + 2y^2 - 7y^3 = 0$$

$$7y^3 - 2y^2 + 8y - 24 = 0$$

(B)

$$6, \quad z = 1 + i$$

$$iz = i - 1$$



$$\arg iz = \frac{3\pi}{4}$$

(B)

$$7, \int x \sin(x^2 + 3) dx.$$

$$= -\frac{1}{2} \cos(x^2 + 3) + C$$

(A)

8, Roots in conjugate pairs since coefficients real. $\therefore 3$

(C)

$$9, \int_0^1 \frac{e^x}{1+e^{2x}}$$

$$= \left[\ln(1+e^x) \right]_0^1$$

$$= \ln(1+e) - \ln 2$$

$$= \ln \left(\frac{1+e}{2} \right)$$

(C)

10

(A)

EXT 2 trial mark breakdown 2014 NAME**COMPLEX NUMBERS**

question	mark
1	/1
2	/1
6	/1
11ai	/1
11aii	/2
11aiii	/2
11bi	/2
11bii	/2
14di	/2
14dii	/1

TOTAL /15**CONICS**

question	mark
3	/1
4	/1
13ai	/3
13aii	/2
13aiii	/2
13aiv	/2
13av	/1

TOTAL /12**GRAPHS**

question	mark
10	/1
14ai	/1
14aii	/1
14aiii	/1
14aiv	/1
14av	/2
16bi	/2
16bii	/3

TOTAL /12**POLYNOMIALS**

question	mark
5	/1
8	/1
11ci	/2
11cii	/2
11ciii	/2
15di	/3
15dii	/2

TOTAL /13**VOLUMES**

question	mark
13bi	/1
13bii	/4
15c	/3

TOTAL /8**INTEGRATION**

question	mark
7	/1
9	/1
12ai	/1
12aii	/2
12b	/3
12ci	/2
12cii	/2
12di	/3
12dii	/2
16a	/3

TOTAL /20**HARDER 3U**

question	mark
14bi	/2
14bii	/2
14c	/2
15a	/4
15b	/3
16ci	/3
16cii	/4

TOTAL /20**SUMMARY**

COMPLEX NUMBERS	/15
CONICS	/12
GRAPHS	/12
POLYNOMIALS	/13
VOLUMES	/8
HARDER 3U	/20
INTEGRATION	/20
TOTAL	/100